GRE Quantitative Reasoning

모의고사 Set 2 — 문제지

시범 세트 · S2 상향

구성
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

대소비교(Quantitative Comparison) 선택지 — 모든 QC 문제 공통
(A) Quantity A is greater. (B) Quantity B is greater.
(C) The two quantities are equal. (D) The relationship cannot be determined from the information given.

안내

- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 2 해설지)에 있습니다.

답안 기록표

S1	1	2	3	4	5	6	7	8	9	10	11	12
답												

S2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
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Section 1 12 Questions · 21 Minutes

Question 1 대소비교

680. Let S_8 represents the set of integers from 1 to 8, and let S_9 represents the set of integers from 1 to 9.
Quantity A
The number of the subsets of S_8
Quantity B
The number of the subsets of S_9

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

1244. Set S consists of all three-digit positive integers that contain only the digits 1, 2, 3, 4, or 5.
Quantity A: The number of integers in S in which all three digits are different
Quantity B: 60

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

599. One of the solutions of the equation $x^2+bx+c=0$ is -3.
Quantity A: 3b
Quantity B: c+9

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

772. Quantity A: The area of a circular region with circumference 10
Quantity B: 9

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 객관식

448. $(60!-59!)/58!=$
A $(59)(58)$
B $(60)(59)$
C $(58)^2$
D $(59)^2$
E $(60)^2$

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Questions 6 to 8 are based on the following data.

Kim's Budget for June

	Budgeted	Actual
Work-Related		
Expenditures		
Transportation	$120	$120
Tolls and Parking	100	100
Uniforms	75	40
Laundry/Dry Cleaning	55	40
Lunch and Snacks	50	80
Non-Work-Related		
Expenditures		
Rent	$800	$800
Automobile	550	500
Savings	300	300
Entertainment/Restaurants	300	380
Groceries	250	290
Heat/Light/Telephone	160	150
Insurance	100	100
Clothing	50	30
Other	90	70

Question 6 객관식

10. Of the following fractions, which best approximates the ratio of total budgeted work-related expenditures to the total amount budgeted?
    A. 1/3
    B. 1/4
    C. 1/5
    D. 1/8
    E. 1/10

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Question 7 객관식

11. If 23.75 percent of actual work-related expenditures for lunch and snacks was for soft drinks, what percent of total actual work-related expenditures was for soft drinks ?

A. 5%
B. 4%
C. 3%
D. 2%
E. 1%

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Question 8 객관식

12. For how many of the 14 categories shown did actual expenditures differ from budgeted expenditures by more than 20 percent of budgeted expenditures ?

A. Three
B. Five
C. Six
D. Seven
E. Nine

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Question 9 객관식

540. An engineer draws two concentric circles on a blueprint. If the diameter of the outer circle is 6 centimeter greater than the diameter of the inner circle, how much greater, in centimeters, is the circumference of the outer circle than the circumference of the inner circle?
A3
B6
C3π
D4π
E6π

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 10 주관식

707. A large cube consists of 216 small identical cubes. What is the number of small cubes that do not have a face that is part of a face of the large cube?

Question 11 복수선택

872. In a certain town, the median daily noon temperature for a 7-day period was less than 60°F. In the town, which of the following could be the total number of days during the 7-day period on which the daily noon temperature was less than 60°F? Indicate all such numbers.

A1
B2
C3
D4
E5
F6
G7

Indicate all that apply. A B C D E F G

Question 12 객관식

759. Three printers, X1, X2 and X3, work only at their respective constant rates. Working together,X1, X2 and X3 can complete a certain job in 9 hours; working together, X2 and X3 can complete the same job in 12 hours. Working alone, how many hours will it take X1 to complete the job?

A12
B15
C18
D24
E36

A B C D E

Section 2 15 Questions · 26 Minutes

Question 1 대소비교

1174. Thirty percent of the members of Club A are also members of Club B. Forty percent of the members of Club B are also members of Club A.
Quantity A: The total number of members of Club A
Quantity B: The total number of members of Club B

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

190. An urn contains 1 red disc, 4 purple discs, and 95 orange discs. Two discs are to be chosen from the urn, simultaneously and at random.
Quantity A: The probability that one of the discs chosen will be red
Quantity B: The probability that both discs chosen will be purple

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교



807. S=1.2, 0.9, 1.3, 1.5
T=1.2x, 0.9x, 1.3x, 1.5x x is a positive number
Quantity A: he standard deviation of the numbers in set S
Quantity B: he standard deviation of the numbers in set T

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

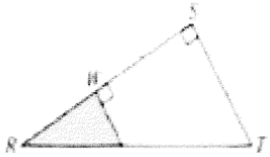
Question 4 대소비교

1214. a, b, c, and d are numbers that are greater than 0 and less than 1.
Quantity A: (a+b)(c+d)
Quantity B: a+b(c+d)

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 대소비교

1253.



The area of the shaded triangular region is 1/5 of the area of the triangular region RST
Quantity A: RW
Quantity B: 1/5(RS)

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 6 객관식

941. The function f is defined for all integers n greater than 2 by f(n)=(1)(2)(3)...(n−2). That is, f(n) is the product of the first n−2 positive integers. What is the value of f(9)/f(8)?
A9
B8
C7
D8/7
E7/6

(A) (B) (C) (D) (E)

Question 7 객관식

508. Last season, Pat won 60% of the chess games she played, and tied 25% as many games as she won. If she tied a total of 3 games, how many more games did she win than lose?
A5
B6
C7
D8
E9

(A) (B) (C) (D) (E)

Question 8 객관식

961. A dealer's gross profit on the sale of a car is defined as the selling price of the car minus its cost to the dealer. A car dealer sold n cars for s dollars each. If the dealer's total gross profit was p dollars, what was the dealer's average (arithmetic mean) cost per car, in dollars?
A (s−p)/n
B s−(p/n)
C (s/n)−p
D p−(s/n)
E sn/p

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Question 9 복수선택

788. Each day for 25 days, the number of people who used a certain library that day was recorded. The standard deviation of the 25 recorded numbers is 12.
Which of the following statements individually provide(s) sufficient additional information to determine the mean of the 25 recorded numbers?
Indicate all such statements.
AThe median of the recorded numbers is 49.
BOne of the recorded numbers, 45, is 1/3 standard deviation below the mean.
CMore than 50 percent of the recorded numbers are between 37 and 61.

Indicate all that apply. [A] [B] [C]

Question 10 주관식

756. During each run of a computer simulation, either the letter X or the letter Y is displayed. For each run of the simulation, if the letter X is displayed, then the probability that X will be displayed in the next run is 0.3. Also for each run of the simulation, if the letter Y is displayed, then the probability that Y will be displayed in the next run is 0.4.
In 7 consecutive runs of the simulation, if X is displayed in the 5th run, what is the probability that X will be displayed in the 7th run?

[ ]

Questions 11 to 13 are based on the following data.

Weekly Rental Rates for Realtor M's Summer Houses on Island X

Location	Number of Bedrooms in House		
	2 Bedrooms	3 Bedrooms	4 Bedrooms
Oceanside	$800	$900	$1,000
Bayside	$600	$700	$800
Inland	$500	$600	$700

Question 11 객관식

14. Of Realtor M's 3-bedroom summer houses on Island X, 5 are oceanside, 6 are bayside, and 1 is inland. What is the average (arithmetic mean) weekly rental rate for these 12 houses?

A. $700
B. $725
C. $733
D. $766
E. $775

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Question 12 객관식

15. Realtor M has a special rate structure for an 8-week rental of a summer house: For each of the 1st and 2nd weeks, the rental charge is the listed weekly rate; for each of the 3rd and 4th weeks, the rental charge is x dollars less than the listed weekly rate; and for each of the remaining weeks, the rental charge is 2x dollars less than the listed weekly rate. In terms of x, what is the total rental charge, in dollars, for an 8-week rental of an oceanside 4-bedroom summer house?

A. 8,000 – 16x
B. 8,000 – 14x
C. 8,000 – 12x
D. 8,000 – 10x
E. 8,000 – 8x

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Question 13 객관식

16. If Realtor M's inland summer houses on Island X generate a total weekly rental income of $8,800 when they are all rented, what is the greatest possible number of inland 4-bedroom summer houses that Realtor M can have?

A. 8
B. 9
C. 10
D. 11
E. 12

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Question 14 객관식

1153. In △ABC, point D lies between A and C, and BD is an altitude. The degree measure of which of the angles of △ABC, if any, could be greater than 90?
A.A only
B.B only
C.C only
D.None of the three angles
E.Any of the three angles

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Question 15 복수선택

857. In the ceiling of a room, an opening was cut in the shape of a square with sides that are 1 foot in length. A circular fixture will be placed over the opening. If the circular fixture covers the square opening completely, which of the following could be the diameter of the circular fixture, in inches? (Note: 1 foot = 12 inches) Indicate all such diameters.
A12
B13
C14
D15
E16
F17
G18

Indicate all that apply. [A] [B] [C] [D] [E] [F] [G]
